

Adem Batur

Computer Engineering Student — AI · Computer Vision · Mobile Development

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PROFILE

Third-year Computer Engineering student at Sakarya University (GPA 3.48/4.00), also completing a minor in Control and Automation. I build end-to-end computer vision systems with Python and PyTorch, and Flutter mobile apps with AI coding agents. Finalist in TEKNOFEST 2026 AI in Aviation and principal investigator of a TÜBİTAK 2209-B funded project. Looking for a compulsory or part-time internship in software development, AI, mobile or data.

EDUCATION

Sakarya University — B.Sc. Computer Engineering 2024 – 2028 (expected) · GPA 3.48 / 4.00
High Honor List (Fall 2024–25, Fall 2025–26) · Honor List (Spring 2024–25, Spring 2025–26)

Sakarya University — Minor in Electrical & Electronics Engineering: Control and Automation 2026 – present
Relevant courses: Data Structures · Database Management Systems · Introduction to Programming (C++) · Web Technologies · Automatic Control

COMPETITIONS AND AWARDS

- TEKNOFEST 2026 AI in Aviation — Finalist (2026).** Captain of the two-person team NURON; I selected the models and wrote the algorithms. Finished the online simulation round, which drew 150+ teams, in 2nd place and reached the on-site finals.
- TÜBİTAK 2209-B Industry-Oriented Undergraduate Research Grant — funded (2026).** Principal investigator of “Ghost Cursor”, an AI agent that operates multiple screens from voice commands (Whisper speech recognition, LLM action planning, UI automation). Advisor: Prof. Dr. Cemil Öz · industry advisor: Datazza.
- TEKNOFEST 2026 5G & AI Smart Road Safety (2026) —** preliminary design report 84 points (threshold 64); scored 103.29 in the second stage, 1.71 points short of the 105 threshold, and did not reach the finals.

PROJECTS

NURON — Perception and GPS-Denied Positioning for UAV Imagery 2025 – 2026
Python · PyTorch · YOLO26 · DPVO · SAM 2.1 · DINOv3 · github.com/adembtr/nuron-teknofest-2026

- Solves object detection, GPS-denied position estimation and reference-object matching on one frame stream, on a single 8 GB GPU, at 0.52–0.86 s per frame (22–36% of the time budget).
- On a local replica of the competition server: ~1.6 m horizontal error over the 1,800 frames after GPS is cut; 98.8% precision in reference matching; vehicle-motion accuracy raised from 0.946 to 0.985.
- Traced failures to their root cause: making the inter-process frame handoff atomic cut corrupt frames from 9% to 0%, and a calibration confidence gate cut the worst session’s error from 982 m to 6.4 m.

Dotnote — Note-Taking App for Tablets and Phones 2026
Flutter · Dart · Riverpod · Drift (SQLite) · go_router · Android / Web · Claude Code · elifapps.com/apps/dotnote

- S Pen handwriting, audio/video placed on the page, A4 pages, PDF annotation, PDF/DOCX export, reminders, home-screen widgets and Google Drive backup; 10 languages including right-to-left Arabic.
- Built by directing Claude Code agents. I owned the requirements, a competitor analysis of 9 apps, the architecture (a shared package reused across apps; Screen → Provider → Repository → DAO) and on-device testing.
- 700+ automated tests, 110+ builds tested on a tablet. I use it daily for my lecture notes; a Google Play release is planned.

Smart Road Safety and Turkish Licence-Plate Recognition 2026
YOLO26-OBb · EfficientNet · fast-plate-ocr · ONNX · github.com/adembtr/turkish-plate-ocr

- End-to-end pipeline around a single-pass vehicle, plate and driver detector: plate deskewing and reading, vehicle type/colour and driver-behaviour classification.
- Benchmarked three OCR engines on the same 95 plates and picked the best (EasyOCR 10, PaddleOCR 52, fast-plate-ocr 74 correct; ~455 plates/s on CPU).

Other Projects 28 public repositories · github.com/adembtr

- ARC-AGI-3 agents — learn rules from their own actions, with no LLM or dataset; cleared 7 levels on 25 games (random agent: 2).
- Library Management System — book, member and loan management with a reporting dashboard (PostgreSQL, Streamlit, Plotly; DBMS course).
- MathTransformer (~0.84M-parameter transformer learning arithmetic from scratch) · RAG document Q&A (ChromaDB) · n8n video automation · VocabForge (JavaScript vocabulary app) · websites for small businesses (SEO, Netlify).

SKILLS AND LANGUAGES

Programming: Python, C++, Dart, JavaScript, SQL, C, C#, Java

AI / Vision: PyTorch, OpenCV, Ultralytics YOLO, SAM 2, DINOv3, visual odometry (DPVO), OCR, RAG, LLM integration

Mobile, Web, Data: Flutter, Riverpod, HTML/CSS/JavaScript, Streamlit · PostgreSQL, SQLite, ChromaDB, FAISS

Tools: Git/GitHub, Linux, CUDA, Conda, Android SDK / adb, n8n, LaTeX, Claude Code

Languages: Turkish (native) · English (professional working proficiency)